SECURE CI/CD PIPELINE FOR A LOAN APPLICATION TRACKING PORTAL

Automated DevOps Architecture • Dual-Role (Borrower/Officer) RBAC • SBI Underwriting Engine • 9-Journey Selenium Quality Gate

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RELEASE / BUILD
v2.1.0-auth (27 Tests)

✓ **CI/CD QUALITY GATE PASSED**

PROBLEM STATEMENT & SCOPE

OVERVIEW

Traditional retail lending workflows suffer from opaque processing, lengthy turnaround times, lack of real-time auditability, and no separation between customer submission and underwriting decisioning.

> **Core Objective:** Deliver an enterprise-grade financial tracking portal powered by a fully automated **Jenkins CI/CD delivery pipeline**, resilient **multi-stage Docker packaging**, and **sub-15s Ansible disaster recovery**.

THE 4 SBI UNDERWRITING PILLARS

FINTECH ENGINE

Automated credit risk assessment evaluating retail borrower solvency during live intake:

SBI Pillar	Threshold Criteria	Underwriting Impact
1. CIBIL Score	≥ 750 (Min 650)	Credit bureau solvency gate
2. Net Income	≥ ₹20,000 / month	Repayment capacity check
3. Applicant Age	21 – 60 Years	SBI retail lending boundary
4. Max FOIR	≤ 40% (Max 50%)	Fixed Obligation to Income Ratio

```
[Borrower Submission] ──►                                  ├─ CIBIL ≥ 750 & FOIR ≤ 40% ──► ✅ PRE_APPROVED (Fast-Track)
                                                           ├─ CIBIL < 650 or FOIR > 50% ──► ❌ HIGH_RISK_EXCEPTION (Alerts)
                                                           └─ Other criteria met ──────────► ⚠ MANUAL_REVIEW (Underwriting)
```

> **Formula:** `FOIR = (Existing EMIs + Proposed EMI) / Net Income × 100`

DUAL-ROLE PORTAL ARCHITECTURE

SECURITY • RBAC

Strict separation between public borrower intake and internal banking operations:

End User (Borrower Portal - `/my-applications`):

- Self-registration & login with personal credentials.
- Auto-filled intake form (Name, Email, Phone from profile).
- **My Applications:** View and track personal loan status only.
- **Read-Only Tracking:** Stage advancement panel is completely hidden from borrowers to prevent privilege escalation.

Administrator (Officer Console - `/dashboard`):

- Global portfolio overview of all bank loan applications.
- Real-time 4-KPI counters & global search by ID or name.
- **Stage Advancement:** KYC Verified → Doc Verified → Underwriting → Approved / Rejected with audit remarks.
- **Alerts Queue (`/alerts`):** Dedicated high-priority review.

ROUTE SECURITY & TAB ISOLATION

ZERO-TRUST

All internal endpoints are guarded; unauthenticated access is blocked:

Route	Permitted	Unauthenticated Policy
`/login`, `/register`	Public	Zero internal tabs shown
`/dashboard`	`ADMIN`	Redirect → `/login`
`/my-applications`	`USER`	Redirect → `/login`
`/apply`	`USER/ADMIN`	Redirect → `/login`
`/loans/{id}`	Owner / Admin	Redirect → `/login`
`/alerts`	`ADMIN` only	Redirect → `/login` (403 for User)

> **Guaranteed Privacy:** Before logging in, navigation tabs (Dashboard, New Application, Alerts) and loan tables are completely hidden from the DOM.

JENKINS DECLARATIVE PIPELINE

PIPELINE AS CODE

Automated 7-stage CI/CD execution triggered via webhook/polling on Port 8080:

```
STAGE 1 (Git main / feature branch) STAGE 2
(18 JUnit 5 Tests) STAGE 3
(9 Chrome Journeys) STAGE 4
(Lightweight image) STAGE 5
(Port 8081 container) STAGE 6
(HTTP 200 OK)
```

> **Strict Quality Gate:** If any unit test or Selenium journey fails, the build halts immediately, captures failure screenshots, and prevents deployment.

SELENIUM HEADLESS QUALITY GATE

9 JOURNEYS • ZERO DEFECT

Full automated browser verification executing in Headless Chrome:

- **J1 (Intake):** Submits borrower form → verifies `LOAN-XXXX` on dashboard.
- **J2 (Search):** Filters live table by applicant name with instant reset.
- **J3 (Drill-Down):** Advances status to `DOCUMENT_VERIFIED` → audits event.
- **J4 (Alerts):** Rejects loan with reason code → verifies alerts queue.
- **J5 (SBI Eligible):** CIBIL=820, FOIR ~13% → tags `PRE_APPROVED`.
- **J6 (SBI Ineligible):** CIBIL=580 → auto-tags `HIGH_RISK_EXCEPTION`.
- **J7 (End-User Portal):** Logs in as applicant → personal tracking → verifies read-only stage view.
- **J8 (Admin Management):** Logs in as officer → advances stage to `APPROVED` → logs remarks → accesses alerts.
- **J9 (Route Protection):** Asserts unauthenticated visitor is blocked from dashboard/alerts and no tabs are shown.

> **Test Execution Summary: 27 Passed / 0 Failed** across unit & UI suites.

LIVE PORTFOLIO METRICS

ACTUATOR • TELEMETRY

27	100%	<15s	0
TESTS PASSED	GATE PASS	ROLLBACK MTTR	FAILURES

Actuator Telemetry: Endpoint `/actuator/health` returns `{"status":"UP"}` ensuring container readiness before traffic cutover.

DOCKER MULTI-STAGE & ISOLATION

CONTAINERIZATION

Hardened container build with isolated process ports:

- **Stage 1 (Builder):** Maven 3.9 + Temurin JDK 17 compiles JAR.
- **Stage 2 (Runtime):** Minimal Eclipse-Temurin 17 JRE Alpine.
- **Security:** Non-root system user (`appuser:appgroup`).

Port	Protocol	Service Assignment
8080	HTTP	Jenkins Controller (CI/CD)
8081	HTTP	Loan Application Runtime
22	TCP	SSH / Ansible Automation

ANSIBLE IAC & DISASTER RECOVERY

SUB-15S ROLLBACK

Idempotent Server Provisioning: Ansible playbook configures system prerequisites, directories, and dependencies with zero configuration drift (`changed=0` on rerun).

> **Automated Rollback (Task 14 Deliverable):** Restores previous stable container release (`loan-portal:v1.0.0`) in under 15 seconds if health check fails:
> `powershell -File scripts\verify_reliability.ps1`

> **Demo Logins:**
> • **Borrower:** `user` / `user123`
> • **Officer:** `admin` / `admin123`